SignalRHub Project Documentation

Real-time notification system | .NET 10 | Version 1.0 | October 2026

About this document

This document explains SignalRHub in simple language. It has three parts. Read the part that matches your role.

Part	Who it is for	What you will find
Part 1: Overview	Everyone	What the project is, what it can do, and the key words
Part 2: User Guide	People who use the notification console	Step-by-step instructions and fixes for common problems
Part 3: Technical Guide	Developers and testers	How it is built, how to run it, settings, reference tables, and a checklist for going live

> **Tip:** New to the project? Read Part 1 first, then go to the part for your role.

Part 1: Overview

1.1 What is SignalRHub?

SignalRHub sends messages to people's browsers instantly. Nobody needs to refresh the page or ask for updates. When something happens, the message simply appears.

Think of a post office with noticeboards. An administrator hands a letter to the clerk. The clerk files a copy in the record book, then a delivery team rushes the letter to the right mailboxes at once. In SignalRHub:

- The clerk is the web service (the API).
- The record book is the database.
- The delivery team is SignalR, the technology that keeps a live line open to each browser.

- Each open browser tab is a mailbox.
- A room is a noticeboard that people can subscribe to.

1.2 What can it do?

Feature	What it means
Live delivery	Messages appear in the browser within a moment.
Three audiences	Send to everyone, to one person, or to a room.
Four severity levels	Info, Success, Warning and Critical show how important a message is.
Saved history	Every notification is stored in the database before it is sent.
Automatic reconnect	If the connection drops, the page reconnects and rejoins its rooms.
Status pings	A background job sends a health message to the system-status room every 30 seconds.
Secure access	Users log in with a token. Only Admins can publish.

1.3 Key words

Word	Simple meaning
Notification	One message with a title, text and severity.
Connection	The live line between one browser tab and the server.
User	A person, identified by a user id such as alice.
Room	A named group, such as room-101. Only people who joined receive its messages.
Hub	The server's meeting point where browsers connect.
Token	A digital pass that proves who you are.

Word	Simple meaning
Role	A permission level. The Admin role can publish.
Publish	Create a notification and send it.
Real time	Delivered immediately, without refreshing.

1.4 How it works in four steps

Step	What happens	Post office comparison
1	An admin presses Publish.	A letter is handed to the clerk.
2	The server checks the pass and saves the message.	The clerk checks ID and files a copy.
3	The server sends it to the chosen mailboxes.	The delivery team runs it over.
4	The browser shows the message.	The letter lands in the mailbox.

> **Note:** The message is saved before it is sent. If saving fails, nothing is sent and the admin sees an error.

Part 2: User Guide

2.1 Before you start

You need:

- The SignalRHub service running. Your developer or administrator starts it.
- A web browser such as Chrome, Edge or Firefox.
- The console address. On a developer computer it is https://localhost:7116/test.html. The port number can differ on your machine.

> **Tip:** To see delivery, open the console in two browser tabs and use a different user name in each tab.

2.2 The screen at a glance

Area	What it does
Status badge (top right)	Shows Offline, Connecting, Live as your name, or Reconnecting.
Signal line	A moving line. It spikes up when you receive a message and down when you send one.
Step 1: Connect	Type a user name and press Connect.
Step 2: Join a room	Subscribe to a room to receive its messages.
Step 3: Publish	Write and send a notification.
Incoming tab	Shows every notification this tab received.
Activity log tab	Shows technical events such as connected, joined or errors.
Counter	Shows how many messages were received and sent, and how many rooms you joined.

2.3 Receive your first notification

Use two tabs, called Tab A and Tab B.

1. Open the console in Tab A. Type alice in User id and press Connect. The badge changes to Live as alice.
2. Open the console in Tab B. Type bob in User id and press Connect.
3. In both tabs, type room-101 in the room box and press Join. A room chip appears under the box.
4. In Tab A, under Publish, choose A room, type room-101, and write a title and a message.
5. Press Publish.
6. Look at both tabs. Each one shows a popup and a new card in the Incoming tab.

> **Note:** You can publish only after you connect. The Join and Publish buttons stay grey until the badge says Live.

2.4 Choosing who receives a message

Send to	Who gets it	What to type in the audience box	Example
Everyone	Every connected tab	Nothing. The box is disabled.	A maintenance notice
One user	All tabs of that user	The user id	bob
A room	Tabs that joined that room	The room id	room-101

> **Tip:** User ids must match exactly. The user id bobs is not the same as bob.

2.5 Severity levels

Level	Use it for	Color
Info	General news	Blue
Success	Something finished well	Green
Warning	Needs attention soon	Amber
Critical	Needs attention now	Red

2.6 What the status badge means

Badge	Meaning	What you do
Offline	Not connected	Press Connect.
Connecting	Opening the connection	Wait a moment.
Live as name	Connected and ready	Join rooms and publish.
Reconnecting	The line dropped and the page is retrying	Wait. It retries by itself.

> **Note:** After a reconnect the page rejoins your rooms automatically. Messages sent while you were offline are not shown again on screen, but they are stored in the database.

2.7 Common problems

What you see	Likely reason	What to do
Join and Publish are grey	You are not connected	Press Connect and wait for Live.
Publish shows HTTP 403	Your pass does not have the Admin role	Ask your administrator. The test page asks for Admin automatically.
Publish shows HTTP 500	The server could not save the message	Tell your developer. See section 3.15.
A message arrives in one tab only	The other tab is not in the room, or the user id is spelled differently	Check the room name and user id.
Nothing arrives from the background job	You did not join system-status	Join the room system-status and wait up to 30 seconds.
Invalid room id	The room name has a space or symbol	Use letters, digits, dash or underscore, up to 64 characters.

Part 3: Technical Guide

3.1 Technology used

Area	Technology
Framework	ASP.NET Core Web API on .NET 10
Live messaging	SignalR over WebSockets
Database	SQL Server with Entity Framework Core

Area	Technology
Login	JWT bearer tokens
Design style	Clean Architecture with four layers
Test page	Plain HTML and JavaScript in wwwroot

3.2 Project layers

The code is split into four layers so each part has one job. One simple rule keeps it tidy: inner layers never know about outer layers.

Layer	Its job	Main items
Domain	The business rules	Notification, NotificationSeverity
Application	The use cases and the interfaces they need	PublishNotificationHandler, INotificationService, NotificationDto
Infrastructure	Database, connection memory and the background job	AppDbContext, NotificationRepository, SystemStatusWorker
API	Entry points and the SignalR adapter	NotificationsController, NotificationHub, SignalRNotificationService, Program.cs

> **Note:** The Application layer asks for a live push through an interface called INotificationService. The API layer supplies the real SignalR version at startup. This keeps business code free of SignalR details and easy to test.

3.3 Folder layout

```
src/
  SignalRHub.Domain/          Entities, enums, events
  SignalRHub.Application/     Handler, interfaces, DTOs, group names
  SignalRHub.Infrastructure/  Database, tracker, background worker
  SignalRHub.Api/             Controllers, hub, extensions, wwwroot/test.html
```

3.4 How a notification travels

This is the path when an admin publishes through the API.

Step	Where	What happens
1	Browser	The admin sends POST /api/notifications with a token.
2	API	The token is checked and the Admin role is confirmed.
3	API	The request body is checked: title, message and audience.
4	Application	PublishNotificationHandler decides the audience.
5	Domain	Notification.Create builds the notification and checks the rules.
6	Infrastructure	The repository saves it to the database.
7	Application	The handler calls INotificationService to push it.
8	API	SignalRNotificationService sends it through SignalR.
9	Browser	Each target tab receives ReceiveNotification and shows it.
10	API	The API replies 202 Accepted to the admin.

> **Note:** Steps 9 and 10 happen at about the same time. A 202 reply means saved and handed to SignalR. It does not mean every screen has finished showing the message.

3.5 How a user connects

1. The browser gets a token. In development it comes from the test endpoint /dev/token.
2. The browser opens a live connection to /hubs/notifications and sends the token.
3. The server checks the token. Without a valid token the connection is refused.
4. The server reads the user id from the token and records the connection.
5. The browser calls JoinRoom for each room it wants.

3.6 Background worker

SystemStatusWorker starts with the application. Every 30 seconds it sends a System status message to the room system-status. It does not save to the database. If one send fails it is logged, and the next one still runs.

3.7 Setup and run

Requirements: the .NET 10 SDK, SQL Server (or LocalDB) and the dotnet-ef tool.

1. Open the solution and restore packages.
2. Fill in appsettings.Development.json (see section 3.8).
3. Create the database table with the commands below, run from the solution folder.
4. Start the Api project.
5. Open https://localhost:7116/test.html, using the port shown in your console.

```
dotnet ef migrations add InitialCreate --project
src/SignalRHub.Infrastructure --startup-project src/SignalRHub.Api
dotnet ef database update --project src/SignalRHub.Infrastructure --startup-
project src/SignalRHub.Api
```

> **Tip:** If your computer does not trust the development certificate, run dotnet dev-certs https --trust once.

3.8 Settings

Setting	What it controls	Example
ConnectionStrings:Default	Database address	Server=localhost;Database=SignalRHub;Trusted_Connection=True;TrustServerCertificate=True
Jwt:Issuer	Who creates tokens	notification-api
Jwt:Audience	Who tokens are for	notification-clients
Jwt:Key	Secret used to sign tokens, 32 characters or more	A long random value
Cors:AllowedOrigins	Websites allowed to connect	http://localhost:4200

```
{
  "ConnectionStrings": { "Default":
"Server=localhost;Database=SignalRHub;Trusted_Connection=True;TrustServerCert
ificate=True" },
  "Jwt": {
    "Issuer": "notification-api",
    "Audience": "notification-clients",
    "Key": "replace-with-a-long-random-secret-at-least-32-chars"
  },
  "Cors": { "AllowedOrigins": [ "http://localhost:4200" ] }
}
```

> **Note:** The app refuses to start if the Jwt or Cors settings are missing or too short. This catches mistakes early.

> **Warning:** Never keep the real Jwt:Key in source control. Use user-secrets or a secrets vault.

3.9 Web API reference

POST /api/notifications

Saves and sends a notification. Requires login and the Admin role.

Field	Required	Description
title	Yes	Headline, up to 200 characters
message	Yes	Body text, up to 2000 characters
severity	Yes	Info, Success, Warning or Critical
target	Yes	All, User or Room
targetValue	When target is User or Room	The user id or room id

Example request body:

```
{
  "title": "Deploy finished",
  "message": "Version 2.4 is live",
  "severity": "Success",
  "target": "Room",
  "targetValue": "room-101"
}
```

Reply	Meaning
202 Accepted	Saved and sent. The reply contains the notification.
400 Bad Request	A field is missing or too long, or targetValue is empty.
401 Unauthorized	The token is missing, wrong or expired.
403 Forbidden	The token is valid but the user is not an Admin.
500 Server error	Saving failed, for example the table does not exist.

Other endpoints

Endpoint	What it does	Who can use it
GET /api/notifications/connections	Returns how many connections are open now	Any logged-in user
GET /dev/token?userId=alice&role=Admin	Returns a test token	Anyone, in Development only

3.10 Live connection reference

Connection address: /hubs/notifications (wss://host/hubs/notifications).

Calls the browser can make:

Method	Input	What it does
JoinRoom	roomId	Subscribes the tab to a room.
LeaveRoom	roomId	Unsubscribes the tab from a room.

Events the server sends to the browser:

Event	Details	When
ReceiveNotification	A notification	A message was sent to this tab.
UserJoinedRoom	roomId, userId	Another user joined a room you are in.
UserLeftRoom	roomId, userId	Another user left a room you are in.

A notification contains: id, title, message, severity and createdAtUtc.

> **Note:** Room ids may use letters, digits, dash and underscore, up to 64 characters. Inside the server a room named room-101 becomes the group room:room-101.

3.11 Database table

Column	Type	Notes
Id	Unique id	Primary key
Title	Text (200)	Required
Message	Text (2000)	Required
Severity	Text (20)	Info, Success, Warning or Critical
TargetUserId	Text (128)	Filled when sent to one user
TargetGroup	Text (128)	Filled when sent to a room, for example room:room-101
CreatedAtUtc	Date and time	Always UTC
IsRead	Yes or no	Starts as 0 (unread)

3.12 Security in simple terms

Topic	What it does
Token login	Every request and connection must carry a valid token.
Admin role	Only tokens with the Admin role can publish.
Allowed websites (CORS)	Only listed websites may call the API from a browser.
Room id rules	Blocks odd or oversized room names.
Message size limit	The hub rejects incoming messages over 32 KB.
Safe display	The test page shows message text as plain text, never as HTML.

> **Warning:** The /dev/token endpoint lets anyone become any user. It is switched on only in Development. Delete it, and test.html, before going live.

3.13 Testing checklist

Test	How	Expected result
Connect	Connect as alice and bob in two tabs	Both badges show Live
Status ping	Both join system-status and wait 30 seconds	Both receive System status
Room	Both join room-101, then send to the room	Both receive it
One user	Send to user bob	Only bob receives it
Everyone	Send to everyone	Both receive it
Database	Check the Notifications table	One row for each publish
Reconnect	Stop the server, then start it again	Reconnecting, then Live, and rooms rejoin
No login	Call the hub without a token	401

3.14 Checklist before going live

- Remove wwwroot/test.html and DevelopmentEndpoints.cs.
- Store Jwt:Key in a secrets vault, not in a file.
- List only real website addresses in Cors:AllowedOrigins.
- Replace the dev token with your real login system.
- Run the database migrations on the live database.
- If you run more than one server, add a Redis backplane and a Redis connection tracker (see section 3.16).
- Turn on logging and monitoring.

3.15 Troubleshooting for developers

Problem	Cause	Fix
Publish returns 500 and the log mentions validation metadata on a record	Validation attributes written with the property: prefix on a record	Remove the prefix, for example [Required] string Title
500 with Invalid object name Notifications	The table does not exist	Run the migration commands in section 3.7.
500 with Cannot open database	Wrong connection string, or SQL Server is off	Check ConnectionStrings:Default.
401 when connecting	Token wrong, expired or key mismatch	Get a new token and check the Jwt settings.
403 when publishing	The token has no Admin role	Add role=Admin to the token.
The browser blocks the request	The website is not in the allowed list	Add its exact address to Cors:AllowedOrigins.
One user never receives messages	The user id differs from the token sub value	Use exactly the same text.
Room messages stop after a reconnect	Rooms were not rejoined	The client must call JoinRoom again after reconnecting.
Reconnect gives up	The server was down longer than the retry list	The page keeps retrying every 3 seconds after that.

3.16 Limits and next steps

Topic	Today	Next improvement
Offline users	Live messages are not replayed	Add an endpoint that returns unread notifications when a user connects.
Read status	IsRead exists but is never updated	Add a mark as read call.

Topic	Today	Next improvement
Several servers	Connection memory is kept per server	Use Redis for the tracker and the SignalR backplane.
Domain events	Recorded but not used	Dispatch them, for example to audit logs or email.
Reliability	Save then push. A push failure after saving is not retried	Add an outbox for guaranteed delivery.

Appendix: Quick reference

Item	Value
Test console	https://localhost:7116/test.html
Live connection address	/hubs/notifications
Publish endpoint	POST /api/notifications
Connection count	GET /api/notifications/connections
Dev token (development only)	GET /dev/token?userId=alice&role=Admin
Status ping room	system-status
Status ping interval	30 seconds
Server keep-alive	Every 15 seconds. A silent client is dropped after 30 seconds.
Page retry delays	0, 2, 5, 10, 20, 30, 30 seconds, then every 3 seconds